

Quasi-Orthogonality in High-Dimensional Hilbert Spaces: A Geometric Companion to Environmental Decoherence

Karl Svozil ¹

¹*Institute for Theoretical Physics, TU Wien, Wiedner Hauptstrasse 8-10/136, A-1040 Vienna, Austria**

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The emergence of classicality from unitary quantum mechanics requires the suppression of macroscopic interference and is conventionally modeled dynamically through environment-induced decoherence. We argue that a substantial portion of this story is *kinematic*: the geometry of a high-dimensional Hilbert space alone forces generic vectors to be approximately orthogonal. Combining the exponential packing bound for ε -quasi-orthogonal sets with Lévy’s concentration of measure on the unit sphere, we show that an environment of N particles supports a doubly-exponential supply of ε -quasi-orthogonal pure states, and that two Haar-random vectors have expected modulus-squared overlap $1/d$ with sub-Gaussian tails. We make the kinematic and dynamical inputs explicit: the geometric argument supplies an enormous reservoir of nearly non-interfering branches, but a genuine dynamical assumption—effective typicality on the relevant subspace, generated by conditional Hamiltonians and fast-scrambling chaotic dynamics—is still required to guarantee that physical environmental histories actually populate this reservoir. We then connect the framework to the infinite-tensor-product programme of von Neumann, Hepp, Grangier–Van Den Bossche and Svozil, showing that quasi-orthogonality at finite N is the operational shadow of strict sectorization at $N = \infty$ and presages the transition from Type I to hyperfinite Type II₁ structure. Self-contained proofs of the geometric statements are given in the appendices.

I. INTRODUCTION

The quantum measurement problem asks how the continuous and reversible Schrödinger evolution gives rise to the discrete, irreversible classical world we observe [1, 2]. Operationally, the question reduces to why the off-diagonal interference terms of a system’s reduced density matrix are unobservable on macroscopic scales.

Bell coined “FAPP” (*for all practical purposes*) to express skepticism toward explanations that rest on the practical impossibility of observing macroscopic interference rather than on a fundamental physical mechanism [3]. Modern decoherence theory [1, 4–6] answers this by showing that interaction with a macroscopic environment dynamically drives the off-diagonal terms toward zero in a preferred (pointer) basis selected by the system–environment coupling.

In this Letter we examine a complementary, purely *geometric* contribution to the same suppression. Inspired by recent work in machine learning on the storage of features in high-dimensional embeddings [7, 8], where relaxing exact orthogonality to ε -quasi-orthogonality unlocks an exponential capacity for distinct directions, we transport the same elementary geometry into the Hilbert space of a macroscopic environment. The relevant tools—the exponential packing of quasi-orthogonal unit vectors [9–11] and the concentration of measure on high-dimensional spheres [12, 13]—are classical, but their direct foundational application to the measurement problem is, to our knowledge, underemphasized in the literature.

Our central claim is moderate. Geometry alone guarantees a vast reservoir of nearly non-interfering pure states; this is a kinematic fact about $\mathcal{H}$ that holds in-

dependently of any Hamiltonian. Whether *physical* environmental branches occupy this reservoir is a dynamical question, addressed by the eigenstate thermalization hypothesis (ETH) [14–17] and by results on scrambling and typicality of unitaries generated by chaotic dynamics [18–20]. We make this division of labor explicit, and we deliberately resist the temptation—present in some informal accounts—to elevate the geometric observation into a stand-alone resolution of the measurement problem.

A further conceptual benefit, developed in Sec. V, is that the framework provides an operational, finite- N language for the structural mechanisms long studied in the algebraic and infinite-tensor-product traditions of von Neumann [21], Murray–von Neumann [22], Hepp [23], Grangier–Van Den Bossche [24, 25], and Svozil [26]. What appears as strict sectorization or a Type-I-to-Type-II₁ transition in the $N = \infty$ limit appears in our framework as exponential quasi-orthogonal packing at finite (but very large) N .

II. QUASI-ORTHOGONALITY AND EXPONENTIAL PACKING

In a d -dimensional Hilbert space $\mathcal{H}_d$, the maximum number of mutually orthogonal unit vectors is exactly d . Two unit vectors $|\psi_i\rangle, |\psi_j\rangle$ are called ε -quasi-orthogonal if

$$|\langle\psi_i|\psi_j\rangle|^2 \leq \varepsilon, \quad (1)$$

and a set $\{|\psi_i\rangle\}_{i=1}^N$ is ε -quasi-orthogonal if every pair satisfies this bound. The off-diagonal terms in a reduced

density matrix arising from such environmental records are weighted by $\langle E_j | E_i \rangle$ and are therefore suppressed in modulus by at most $\sqrt{\varepsilon}$.

The cardinality $N_\varepsilon(d)$ of the largest such set scales not linearly but exponentially in d :

$$\exp(c\varepsilon d) \leq N_\varepsilon(d) \leq \exp(C\varepsilon d), \quad (2)$$

for absolute constants $c, C > 0$, valid in the regime $2/d \leq \varepsilon \leq 1/2$. The lower bound follows from a probabilistic construction (Appendix); the upper bound is a Kabatiansky–Levenshtein-type estimate [27–29]. The linear-in- ε scaling in the exponent reflects the squared-overlap convention used in the definition above: an ε -quasi-orthogonal set in this convention is a code of minimum angular distance $\arccos \sqrt{\varepsilon}$, whose spherical-cap volume scales as $(1 - \varepsilon)^{\Theta(d)}$. Had we instead defined quasi-orthogonality through the amplitude threshold $|\langle \psi_i | \psi_j \rangle| \leq \eta$, the corresponding bound would read $\exp(c'\eta^2 d)$, recovering the form familiar from the Johnson–Lindenstrauss flattening lemma [11, 30, 31]; the two conventions are related by $\varepsilon = \eta^2$. Both are manifestations of concentration of measure in high dimensions.

Equation (2) has a striking implication for many-body physics. For a quantum system of N qubits, $d = 2^N$, and the number of available ε -quasi-orthogonal directions scales as

$$N_\varepsilon(2^N) \sim \exp(c\varepsilon 2^N), \quad (3)$$

i.e. *doubly* exponential in the particle number. For a macroscopic environment with $N \sim 10^{23}$, this number exceeds any conceivable bound on the count of distinct macroscopic histories in the observable universe, by an astronomical margin.

III. LÉVY'S LEMMA AND TYPICAL OVERLAPS

The packing bound (2) concerns extremal configurations. The complementary statement, that *generic* (Haar-random) pairs of pure states are already quasi-orthogonal, is a consequence of Lévy's lemma [12, 13].

Identify $\mathcal{H}_d \cong \mathbb{C}^d$ with $\mathbb{R}^{2d}$ and let $\mathbb{S}^{2d-1}$ denote the unit sphere equipped with the uniform (Haar) measure. For an L -Lipschitz function $f : \mathbb{S}^{2d-1} \rightarrow \mathbb{R}$,

$$\mathbb{P}\left(|f(|\psi\rangle) - \mathbb{E}f| \geq \delta\right) \leq 2 \exp\left[-\frac{(2d-1)\delta^2}{9\pi^3 L^2}\right]. \quad (4)$$

Take $f(|\psi\rangle) = |\langle \phi | \psi \rangle|^2$ for a fixed reference state $|\phi\rangle$. Two short calculations (Appendix) yield

$$\mathbb{E}_{|\psi\rangle \sim \mu_{\text{Haar}}} [|\langle \phi | \psi \rangle|^2] = \frac{1}{d}, \quad L \leq 2. \quad (5)$$

Substituting into (4) gives the sub-Gaussian tail bound

$$\mathbb{P}\left(|\langle \phi | \psi \rangle|^2 \geq \frac{1}{d} + \delta\right) \leq 2 \exp\left[-\frac{(2d-1)\delta^2}{36\pi^3}\right]. \quad (6)$$

For this particular observable a sharper and fully explicit statement is available. Standard unitary-invariance arguments show that for fixed $|\phi\rangle$ and Haar-distributed $|\psi\rangle \in \mathbb{S}^{2d-1}$ the random variable $X = |\langle \phi | \psi \rangle|^2$ has the Beta(1, $d-1$) distribution, with explicit survival function

$$\mathbb{P}(|\langle \phi | \psi \rangle|^2 \geq \varepsilon) = (1 - \varepsilon)^{d-1} \leq e^{-(d-1)\varepsilon}. \quad (7)$$

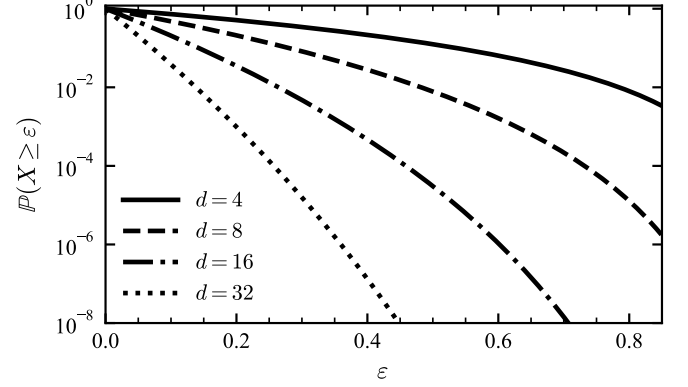


FIG. 1. Exact Haar survival probability for $X = |\langle \phi | \psi \rangle|^2$ in $\mathbb{C}^d$: $\mathbb{P}(X \geq \varepsilon) = (1 - \varepsilon)^{d-1}$. Curves are shown for $d = 4, 8, 16, 32$. The rapid collapse of the right tail illustrates the finite-size onset of quasi-orthogonality.

To make the overlap statistics concrete, consider the smallest nontrivial many-body environment, $N = 3$ qubits, so that $d = 2^N = 8$. Fix $|\phi\rangle = |000\rangle$ and draw $|\psi\rangle$ Haar-randomly from $\mathbb{S}^{15}$. Then

$$X := |\langle \phi | \psi \rangle|^2$$

has the exact Beta(1, 7) density

$$p_8(x) = 7(1 - x)^6, \quad 0 \leq x \leq 1,$$

with mean $\mathbb{E}[X] = 1/8$ and median

$$x_{1/2} = 1 - 2^{-1/7} \approx 9.4 \times 10^{-2}.$$

Even at this modest dimension, large overlaps are already suppressed:

$$\begin{aligned} \mathbb{P}(X \geq 1/4) &= 0.75^7 \approx 1.34 \times 10^{-1}, \\ \mathbb{P}(X \geq 1/2) &= 2^{-7} \approx 7.8 \times 10^{-3}. \end{aligned}$$

For $N = 4$ qubits ($d = 16$), the corresponding probabilities become

$$\begin{aligned} \mathbb{P}(X \geq 1/4) &= 0.75^{15} \approx 1.33 \times 10^{-2}, \\ \mathbb{P}(X \geq 1/2) &= 2^{-15} \approx 3.1 \times 10^{-5}. \end{aligned}$$

A Monte Carlo histogram of Haar-random samples agrees with the analytic density and makes visually clear how rapidly the overlap distribution is pushed toward zero as d increases. The exponent in (7) is linear in ε , whereas (6)

(a Lipschitz consequence of Lévy’s lemma) yields only a quadratic-in- δ Gaussian envelope. For the present application either bound suffices; it is (7), however, that is responsible for the linear-in- ε scaling of the packing exponent in (2), and we use it for the proof of that bound in Appendix . For $d = 2^N$ with N macroscopic, the probability in (7) decays as $\exp[-(2^N - 1)\varepsilon]$, which is doubly exponential in N for any fixed $\varepsilon > 0$: *any two Haar-random pure states in a macroscopic Hilbert space are ε -quasi-orthogonal with overwhelming probability*. For a visual impression of how rapidly the right tail collapses already at modest dimension, see Fig. 1.

IV. APPLICATION TO THE MEASUREMENT PROCESS

Consider a system $\mathcal{S}$ coupled to an apparatus–environment $\mathcal{E}$ initialised in $|E_0\rangle$:

$$|\Psi(0)\rangle = \left(\sum_i c_i |s_i\rangle\right) \otimes |E_0\rangle. \quad (8)$$

A measurement-like unitary correlates pointer outcomes with environmental records:

$$|\Psi(t)\rangle = \sum_i c_i |s_i\rangle \otimes |E_i(t)\rangle. \quad (9)$$

The reduced density matrix on $\mathcal{S}$ is

$$\rho_{\mathcal{S}}(t) = \sum_{i,j} c_i c_j^* \langle E_j(t) | E_i(t) \rangle |s_i\rangle \langle s_j|. \quad (10)$$

Off-diagonal coherences are weighted by the environmental overlaps $\langle E_j(t) | E_i(t) \rangle$.

The kinematic claim.

Throughout this section we adopt natural units $k_B = \hbar = 1$, so that entropies are dimensionless and inverse temperatures have units of time. If the branch states $\{|E_i(t)\rangle\}_{i \neq j}$ are well-modeled as a typical pair on the relevant high-dimensional subspace of $\mathcal{H}_{\mathcal{E}}$, then

$$\mathbb{E}[|\langle E_j(t) | E_i(t) \rangle|^2] = \frac{1}{d_{\text{eff}}}, \quad (11)$$

with d_{eff} the dimension of the subspace explored by the dynamics. For a generic chaotic many-body environment the accessible subspace is the microcanonical shell, so

$$d_{\text{eff}} \sim e^S, \quad S \sim N, \quad (12)$$

with S the (dimensionless) microcanonical entropy of the relevant environmental sector and N the number of constituent degrees of freedom; in conventional units the

right-hand sides of (12) read e^{S/k_B} and Nk_B , respectively. Equation (12) ties the geometric parameter d_{eff} directly to standard thermodynamics: the mean-square overlap is exponential in the entropy of the environment, so the typical overlap amplitude is of order $e^{-S/2}$. This is vastly stronger than needed to render coherences operationally invisible on any timescale shorter than the Poincaré recurrence time, which for a generic chaotic many-body system is astronomically large—heuristic arguments suggest a scaling at least as $\exp(d_{\text{eff}}) \sim \exp(e^S)$, but the precise value is model-dependent and irrelevant to our argument [32, 33].

The dynamical assumption.

The conditional “if” above is doing real work and must not be elided. The branch states $|E_i(t)\rangle$ are not random vectors plucked from $\mathcal{H}_{\mathcal{E}}$; they are deterministic images of $|E_0\rangle$ under specific unitaries $U_i(t)$ generated by the system–environment Hamiltonian.

A natural mechanism, however, makes the typicality model defensible. Generic system–environment couplings take the schematic form

$$H_{\text{int}} = \sum_i |s_i\rangle \langle s_i| \otimes H_{\mathcal{E},i}, \quad (13)$$

i.e. the environment evolves under a *different conditional Hamiltonian* $H_{\mathcal{E},i}$ for each pointer index. When these conditional generators are non-commuting and chaotic—as they generically are for macroscopic environments with local interactions—the unitaries $U_i(t) = e^{-iH_{\mathcal{E},i}t}$ rapidly delocalise the common initial state $|E_0\rangle$ across the microcanonical shell. A standard benchmark for this delocalisation is the fast-scrambling time [18–20],

$$t_* \gtrsim \frac{\beta}{2\pi} \log N, \quad (14)$$

where $\beta = 1/T$ in the natural units adopted above and N is the number of local degrees of freedom. The qualifier “fast” refers precisely to this *logarithmic* dependence on system size: it is to be contrasted with the much slower power-law scrambling characteristic of diffusive or integrable dynamics, in which t_* grows polynomially in N . The bound is saturated, in particular, by black-hole-like and matrix-model dynamics [18–20]. For $t \gtrsim t_*$, each branch state $|E_i(t)\rangle$ is effectively delocalised across the accessible microcanonical shell, and one expects the pair $\{|E_i(t)\rangle, |E_j(t)\rangle\}$ with $i \neq j$ to behave, at the level of low-degree polynomial observables, like a typical pair of points on that shell [34]. We emphasize, however, that this is a *modeling* assumption, not a theorem: the two branch states are deterministic images of a common initial state under different unitaries and are not literally independent Haar-random vectors. The defensible statement is the weaker one that the ensemble of conditional

dynamics, varied over the natural chaotic class to which $H_{\mathcal{E},i}$ belongs, induces overlap statistics that approach those of a typical pair on the accessible shell, with corrections subleading in $1/d_{\text{eff}}$. Under this (standard but non-trivial) assumption, the geometric estimates of Sec. III apply.

The supporting dynamical results—ETH [14–17], Page’s calculation [35], Reimann’s typicality theorems [36, 37]—all converge on the same picture. For integrable, many-body-localised, or weakly-coupled systems these assumptions can fail, and the geometric reservoir, while available, is not occupied by physical histories. The geometric statement is therefore a kinematic *license*, conditional on a dynamical input that is generically but not universally satisfied.

What this does not solve.

Two further caveats are essential and are sometimes blurred in informal presentations.

The preferred-basis problem. Geometric quasi-orthogonality is basis-independent: if $\{|E_i\rangle\}$ are mutually quasi-orthogonal, so is any unitary rotation of them. Decoherence theory’s einselection mechanism [5, 38] singles out a *specific* basis (typically quasi-local in position) determined by the form of H_{int} in Eq. (13). The geometric framework is silent on this selection; it tells us the reservoir is large but not which directions in it are physically realized.

The reality of unselected branches. As Bell stressed, suppression of off-diagonal terms to any nonzero level—no matter how small—does not by itself convert an improper mixture into a proper one [3, 39]. To deny ontological status to non-actualized branches one needs an interpretive commitment (collapse, hidden variables, Everettian branching), not a sharper bound on overlaps. Our geometry sharpens the FAPP suppression dramatically, but FAPP, however small, remains FAPP.

V. CONNECTIONS TO THE LITERATURE

A. Canonical typicality.

The framework here is a near-relative of canonical typicality [36, 37, 40–42]: Lévy’s lemma applied to the partial trace shows that almost every pure state of a large composite system has a reduced state close to the canonical thermal state on a small subsystem. Our application is dual, asking instead about *pairwise* overlaps between distinct macroscopic records.

B. Everettian branching.

The doubly-exponential packing capacity makes precise an intuition often invoked in defenses of the Many-Worlds Interpretation [39, 43]: the Hilbert space “has room” for an unfathomable number of branches without their kinematically interfering. We emphasize, however, that kinematic room is necessary but not sufficient for the Everettian picture, which additionally requires a dynamical branching structure and a probabilistic interpretation [39].

C. Quantum Darwinism.

Zurek’s quantum Darwinism [44, 45] requires the environment to redundantly encode a single classical record across many fragments. The geometric capacity demonstrated here underwrites the very possibility of such redundant encoding: the environment is large enough to host many nearly independent copies, by the same packing bound (2).

D. Random-matrix and chaotic-dynamics approaches.

Recent work has connected typicality to dynamical chaos through random-matrix models of quantum thermalization [17, 46], and through bounds on scrambling rates [20]. Our geometric perspective is consistent with these and clarifies why such approaches succeed: once dynamics drives a state into a typical region of a high-dimensional subspace, the kinematic conclusions of Sec. III take over.

E. Sectorization, factorization, and finite- N shadows.

A complementary—and historically deeper—route to FAPP irreversibility proceeds not through finite-dimensional concentration of measure, but through the structural pathology of *infinite* tensor products. Following von Neumann’s analysis of “incomplete infinite direct products” [21] and its development in algebraic quantum field theory [47–49] and in the statistical-mechanical theory of measurement [23, 50], the infinite tensor product space $\bigotimes_{n=1}^{\infty} \mathcal{H}_n$ decomposes into mutually orthogonal *sectors*: equivalence classes of sequences $\{|\psi_n\rangle\}, \{|\phi_n\rangle\}$ for which

$$\sum_{n=1}^{\infty} (1 - |\langle \psi_n | \phi_n \rangle|) < \infty. \quad (15)$$

Equation (15) is the phase-insensitive form appropriate to macroscopic/operational sectors; von Neumann’s original 1939 condition for strict vector equivalence retains

phase information and reads $\sum_n |1 - \langle \psi_n | \phi_n \rangle| < \infty$ [21], the two coinciding once one quotients by global phases of the individual factors—which is the appropriate identification for sectors interpreted as classical pointer outcomes. Vectors in different sectors are *strictly* orthogonal, and no convergent infinite tensor product of unitaries can interconvert them. Sectorization has been advocated as a structural mechanism for the emergence of classical pointer states by Hepp [23] and, more recently, by Grangier and Van Den Bossche [24, 25]; an extended argument tying nested Wigner’s-friend chains to the same limit appears in Svozil [26].

The geometric perspective developed here can be read as the *finite- N shadow* of this picture. Three correspondences make this precise.

(i) *Quasi-sectors at finite N .* At finite N , the strict equivalence relation defining sectors is replaced by an approximate one at threshold ε : a *quasi-sector* is a maximal ε -quasi-orthogonal family in $\mathcal{H}_{2^N}$. The packing bound (2) yields $N_\varepsilon(2^N) \sim e^{c\varepsilon 2^N}$ such quasi-sectors. This finite- N notion is the shadow of the infinite-tensor-product criterion: for product vectors $|\Psi\rangle = \bigotimes_n |\psi_n\rangle$ and $|\Phi\rangle = \bigotimes_n |\phi_n\rangle$, the condition (15) means that the infinite product of moduli $\prod_n |\langle \psi_n | \phi_n \rangle|$ converges to a nonzero limit and the two vectors belong to the same sector, whereas divergence of the sum forces the product overlap to vanish and places the vectors in different, mutually orthogonal sectors. In this sense the finite- N packing bound is an operational version of sectorization: small per-component mismatches can accumulate into a macroscopic suppression of the total overlap. Equivalently, the central inequality used by Svozil [26] and by Van Den Bossche–Grangier [25]—that small per-component mismatches can compound into a vanishing inner product in the infinite limit—is, at finite N , controlled quantitatively by Eqs. (6) and (7).

(ii) *From Type I to hyperfinite Type II₁.* The local algebra at finite N is the Type I_{2^N} factor $\mathcal{A}_N = \mathcal{M}_{2^N}(\mathbb{C})$, whose minimal projections are rank one. The inductive limit of the algebras $\mathcal{A}_N$ along the inclusions $a \mapsto a \otimes \mathbf{1}$ is the uniformly hyperfinite (UHF) C^* -algebra of type 2^∞ ; in the Gel’fand–Naimark–Segal (GNS) representation of the unique tracial state compatible with these inclusions, its weak operator closure is the hyperfinite Type II₁ factor $\mathcal{R}$ [22, 51], which is *diffuse*: it has no minimal projections, and the trace takes a continuum of values in $[0, 1]$. The doubly-exponential proliferation of ε -quasi-orthogonal projections derived in Sec. II is the kinematic precursor of this diffuseness. At finite N , the rank-one projections are minimal only in the strict orthogonal sense; under the more physical equivalence furnished by quasi-orthogonality, no single projection is operationally distinguishable from a family of $e^{c\varepsilon 2^N}$ near-equivalents. Diffuseness does not appear suddenly at $N = \infty$; it is approached gradually, with the geometric capacity pro-

viding a sharp finite- N measure of proximity to the continuum limit.

For the corresponding Type III structures encountered in algebraic quantum field theory and in recent gravitational settings [52], an analogous shadow exists but is more subtle: Type III factors lack any normalized trace, so the appropriate finite- N proxy is not a projection count but the entanglement-spectrum statistics, which our framework also constrains via Eq. (12).

(iii) *Two routes to FAPP, one mathematical fact.* Svozil [26] and Grangier–Van Den Bossche [24, 25] obtain irreversibility by passing to the strict $N = \infty$ limit and observing that the resulting transitions cease to be unitary in the sense of the original finite-dimensional theory. Our route obtains operational irreversibility at finite (but enormous) N , where unitarity remains mathematically intact but the mean-square size of off-diagonal coherences is suppressed by $\sim e^{-S}$, with the typical amplitude suppressed by $\sim e^{-S/2}$, where $S \sim N$ in natural units (and $S \sim Nk_B$ in conventional ones). The two routes are duals of one another: the infinite-tensor-product argument trades finite control for a clean limit; the geometric argument trades a clean limit for finite control. Both rest on the same mathematical phenomenon—the orthogonality (or near-orthogonality) of generic vectors in product spaces of high cardinality—and both remain vulnerable to the same Bell-style objection that “FAPP is not exact.” Whether one prefers the kinematic shadow or the limit it shadows is, in the end, partly a question of one’s willingness to accept transfinite mathematical objects as physically meaningful.

F. Machine-learning embeddings.

The motivating analogy to high-dimensional vector embeddings in language models and superposition codes [7, 8] is more than rhetorical: the same packing bound that enables a network with d hidden dimensions to store $\sim e^{c\varepsilon d}$ approximately-disentangled features (in the squared-correlation convention adopted here, equivalent to $e^{c'\eta^2 d}$ in the more common amplitude-correlation convention with $\eta = \sqrt{\varepsilon}$) enables a Hilbert space of dimension d to host that many approximately-classical records. Both phenomena are facets of high-dimensional geometry and concentration of measure.

VI. DISCUSSION

The geometric perspective supplies a quantitative kinematic floor on environmental orthogonality but does not, by itself, vindicate Bell’s worry about FAPP. What it does accomplish is to disentangle two distinct contributions to decoherence that are often bundled together. The first is dynamical: the system–environment coupling

(13) generates entanglement, selects a quasi-local pointer basis, and drives initial overlaps to small values on the fast-scrambling timescale (14). The second is geometric: once states reach a typical region of a high-dimensional space, their mean-square overlaps are concentrated near $1/d_{\text{eff}} \sim e^{-S}$, so the typical overlap amplitude is of order $e^{-S/2}$, regardless of how the states were generated. The two contributions are logically independent, and our analysis is a clean separation of variables.

A natural follow-up is to ask whether the geometric reservoir can be put under direct experimental pressure. Mesoscopic interference experiments—in superconducting qubits, trapped ions, or matter-wave interferometers [53]—probe regimes in which d_{eff} is moderate and the typicality assumption is partially violated; deviations from the predicted overlap distribution would constrain how rapidly chaotic dynamics drives states into typicality. We leave this question to future work.

VII. CONCLUSION

We have argued that the suppression of macroscopic interference has a clean kinematic component—the exponential packing and concentration of measure of high-dimensional Hilbert spaces—which complements the dynamical mechanism of environmental decoherence and provides a finite- N , operational shadow of the infinite-tensor-product programme. In a d -dimensional space, an exponential-in- d supply of ε -quasi-orthogonal pure states is available, and Haar-random pairs satisfy $\mathbb{E}|\langle\phi|\psi\rangle|^2 = 1/d$ with the explicit Beta tail $\mathbb{P}(|\langle\phi|\psi\rangle|^2 \geq \varepsilon) = (1 - \varepsilon)^{d-1}$ and the corresponding sub-Gaussian Lévy bound. For macroscopic N , with $d_{\text{eff}} \sim e^S$ and S extensive in N (i.e. $S \sim N$ in natural units, or $S \sim Nk_B$ in conventional ones), this makes the mean-square inter-branch overlap exponentially small in N , while the number of available quasi-orthogonal directions is doubly exponential in N whenever the conditional dynamics of Eq. (13) drive the environment into a typical region of its micro-canonical shell on the fast-scrambling timescale (14). The geometry does not single out a preferred basis or convert improper mixtures into proper ones, and to that extent does not resolve the conceptual core of the measurement problem. It does, however, make precise—and surprisingly extreme—the size of the FAPP cushion on which the classical world rests, and it makes the relationship between quasi-orthogonality at finite N and strict sectorization at $N = \infty$ entirely explicit.

Appendix A: Mean and Lipschitz Bound for Overlap

Throughout this appendix, $\|\cdot\|$ denotes the Euclidean (chordal) norm $\|\psi\rangle = \sqrt{\langle\psi|\psi\rangle}$ on $\mathcal{H}_d \cong \mathbb{C}^d \cong \mathbb{R}^{2d}$, restricted to the unit sphere $\mathbb{S}^{2d-1}$. This is the norm ap-

pearing in Lévy’s lemma (4), and is to be distinguished from the trace, Hilbert–Schmidt, and operator norms used elsewhere in the paper.

Mean. Let $|\phi\rangle \in \mathcal{H}_d$ be fixed and let $|\psi\rangle$ be Haar-distributed on the unit sphere. By unitary invariance, $\mathbb{E}|\langle\phi|\psi\rangle|^2$ depends only on the dimension. Choose any orthonormal basis $\{|e_k\rangle\}_{k=1}^d$. Then

$$1 = \mathbb{E} \sum_{k=1}^d |\langle e_k|\psi\rangle|^2 = d \cdot \mathbb{E}|\langle e_1|\psi\rangle|^2,$$

so $\mathbb{E}|\langle\phi|\psi\rangle|^2 = 1/d$. $\square$

Lipschitz constant. Let $f(|\psi\rangle) = |\langle\phi|\psi\rangle|^2$ on the unit sphere with the chordal Euclidean metric $\|\cdot\|$ defined above. For unit vectors $|\psi_1\rangle, |\psi_2\rangle$,

$$\begin{aligned} |f(|\psi_1\rangle) - f(|\psi_2\rangle)| &= ||\langle\phi|\psi_1\rangle|^2 - |\langle\phi|\psi_2\rangle|^2| \\ &\leq (|\langle\phi|\psi_1\rangle| + |\langle\phi|\psi_2\rangle|) |\langle\phi|\psi_1\rangle - \langle\phi|\psi_2\rangle| \\ &\leq 2 |\langle\phi|\psi_1 - \psi_2\rangle| \leq 2 \|\psi_1 - \psi_2\|, \end{aligned}$$

where we used $|\langle\phi|\psi_k\rangle| \leq 1$ and Cauchy–Schwarz. Hence $L \leq 2$. $\square$

Appendix B: Lower Bound on Quasi-Orthogonal Packing

Probabilistic construction. Sample N unit vectors $|\psi_1\rangle, \dots, |\psi_N\rangle \in \mathcal{H}_d$ independently from the Haar measure on $\mathbb{S}^{2d-1}$. For any pair $i \neq j$, by the exact Beta tail (7),

$$\mathbb{P}(|\langle\psi_i|\psi_j\rangle|^2 \geq \varepsilon) = (1 - \varepsilon)^{d-1} \leq e^{-(d-1)\varepsilon}.$$

By a union bound over the $\binom{N}{2} < N^2/2$ pairs,

$$\mathbb{P}(\exists i \neq j : |\langle\psi_i|\psi_j\rangle|^2 \geq \varepsilon) \leq \frac{1}{2} N^2 e^{-(d-1)\varepsilon}.$$

Choosing $N = \lceil e^{(d-1)\varepsilon/2-1/2} \rceil$ makes this bound less than 1, so an ε -quasi-orthogonal set of size N exists. Thus $N_\varepsilon(d) \geq \exp(c\varepsilon d)$ with $c = 1/2$ for d large. $\square$

The matching upper bound $N_\varepsilon(d) \leq \exp(C\varepsilon d)$ follows from sphere-packing arguments of Kabatiansky–Levenshtein type [27, 28]: an ε -quasi-orthogonal set is in particular a code of minimum angular distance $\theta = \arccos \sqrt{\varepsilon}$, whose cardinality is bounded above by the inverse of the normalized volume of a spherical cap of half-angle $\theta/2$. The constants c, C are not identical, but both lie in $(0, \infty)$ uniformly in d for ε in any compact subinterval of $(0, 1)$.

We remark that the corresponding bound for the *amplitude* threshold $|\langle\psi_i|\psi_j\rangle| \leq \eta$ is $\exp(c'\eta^2 d)$ with the well-known Johnson–Lindenstrauss exponent [11, 30]; the two are related by $\varepsilon = \eta^2$, which translates the exponent $\eta^2 d$ into εd as above.

Appendix C: Effective Dimension and the Microcanonical Entropy

In (6) the relevant dimension is the dimension of the subspace $\mathcal{H}_{\text{acc}} \subset \mathcal{H}_{\mathcal{E}}$ accessible to the dynamics from the initial state on the timescale considered, not the full Hilbert space dimension. Energy conservation, conserved charges, and finite-time evolution all restrict the accessible subspace. Concretely, for a many-body system with Hamiltonian $H_{\mathcal{E}}$ and an initial state localized within an energy window $[E, E + \Delta E]$, the accessible subspace is well approximated by the microcanonical shell, and its dimension is

$$d_{\text{eff}} = \dim \mathcal{H}_{\text{acc}} \sim e^{S(E)/k_B} \quad (16)$$

with $S(E)$ the microcanonical entropy. In the natural units adopted in Sec. IV ($k_B = \hbar = 1$), the right-hand side reads $e^{S(E)}$, consistent with (12). For chaotic many-body dynamics with bounded local interactions, $S(E)$ is extensive in N (Boltzmann scaling), so d_{eff} remains exponential in N and the qualitative conclusions of Sec. IV survive.

For integrable or many-body-localized systems, conserved local quantities restrict the dynamics to a polynomial-in- N submanifold of the microcanonical shell [16, 17], and d_{eff} can be polynomial or even constant in N . The geometric reservoir is correspondingly smaller, and the conditional dynamical assumption of Sec. IV is correspondingly violated. This is a further reason to view the geometric statement as conditional on a dynamical input, and provides a quantitative diagnostic: any experimental observation of slower-than- e^{-S} decoherence is direct evidence of obstructions to effective typicality on the accessible subspace.

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* karl.svozil@tuwien.ac.at

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